

Tran Quoc Truong

Applied AI Backend Engineer | Agentic QA & RAG Systems

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Summary

Applied AI Backend Engineer focused on building AI agent workflows, RAG systems, and agent testing/evaluation platforms using Python, Django/DRF, FastAPI, async workers, and vector databases.

Combines QA automation experience in Core Banking and Payment Hub domains with LLM evaluation, traceable retrieval, and production-oriented backend workflows.

Experience

FPT Software, Applied AI Engineer & QA Automation Engineer

Ho Chi Minh City, Vietnam

Dec 2024 – present

1 year 7 months

- Built AI-assisted QA workflows for document understanding, RAG Q&A, test-case generation, duplicate bug detection, traceability, and change impact analysis.
- Designed Python backend orchestration with Django/DRF, FastAPI, Redis/Celery, PostgreSQL, async workers, service layers, and clear domain boundaries.
- Delivered API, Web, and Mobile automation solutions that increased test coverage by 40% and reduced manual testing effort by up to 70%.
- Led an omnichannel Core Banking automation PoC across Web, iOS, and Android, helping secure a ~20 man-month contract.

FPT Software Academy, Full-Stack Developer Intern

Ho Chi Minh City, Vietnam

Sept 2024 – Dec 2024

4 months

- Built Django-based web applications and deployed containerized services with Docker.
- Improved deployment reliability through production-oriented configuration and Cloudflare integration.
- Reduced deployment cycle time by approximately 50% through cleaner containerized delivery workflows.

Projects

Omni-Agent — Django/DRF AI Agent Hub for Knowledge Builder & Agent Testing

A Django/DRF-based AI Agent Hub where the main backend owns product logic, domain state, workflow rules, jobs, artifacts, and APIs, while stateless workers and services execute technical AI tasks.

- Designed Omni-Agent as the main Django/DRF backend for project-centric AI agent products, owning domain logic, permissions, workflow state, job lifecycle, artifact metadata, and product module boundaries.
- Built Knowledge Builder as the first product module, supporting versioned document workflows, OCR/chunking pipelines, S3-compatible artifacts, presigned preview/download URLs, and traceable RAG foundations.
- Modeled external workers and REST services as stateless executors for technical tasks such as OCR, document parsing, chunking, embedding, retrieval, and evaluation, with Django persisting domain results.
- Proposed ATA as an Agent Testing and Audit module for evaluating RAG and LLM-agent workflows through eval datasets, judge/critic/meta-judge patterns, and regression reports.
- Prototyped React/Vite console workflows for project navigation, product switching, document workspace review, version selection, preview tabs, and processing status visibility.

APIT — Agent Programmatic Integration Testing

An LLM-based system that analyzes API documentation and generates structured API test cases.

- Built the core AI agent for API document understanding, structured test-case generation, and automated output evaluation.
- Evaluated LoRA and QLoRA approaches for domain-specific structured test document generation.
- Fine-tuned Qwen-2.5-3B, achieving macro-F1 0.655 and outperforming Llama-3.2-3B by 41.5% in the evaluation setup.
- Implemented FastAPI-based model evaluation workflows and LLM-as-a-Judge experiments for generated output comparison.

Banking QA Automation — Core Banking & Payment Hub

Automation and validation work for Core Banking omnichannel flows and Payment Hub transaction integrations.

- Built cross-platform automation flows with Katalon and Appium for Web, iOS, and Android regression coverage.
- Validated critical business journeys consistently across Web, iOS, and Android under one automation strategy.
- Authored more than 1,500 lines of SQL validation queries to verify post-transaction data consistency across payment integration flows.
- Reduced manual QA effort by 70% through reusable validation scripts, regression scenarios, and repeatable API/database checks.

Skills

Applied AI & Agent Evaluation: RAG, Hybrid RAG, LangGraph, LangChain, LiteLLM, Langfuse, LLM Evaluation, LLM-as-a-Judge, eval datasets, judge/critic/meta-judge patterns

Backend & Systems: Python, Django/DRF, FastAPI, PostgreSQL, Redis/Celery, Docker, REST APIs, async workers, service layers, domain modeling

QA Automation & Banking: Agentic QA, Agent Testing, API Testing, Postman, Katalon, Appium, JMeter, SQL validation, Core Banking, Payment Hub

Frontend Prototyping: React, Vite, Tailwind CSS, internal dashboard UI, AI-assisted development

Education

FPT School of Business & Technology, Master of Software Engineering in AI

Ho Chi Minh City, Vietnam
May 2026 – present

FPT University, Bachelor of Artificial Intelligence

Ho Chi Minh City, Vietnam
2021 – 2025

Languages

Vietnamese: Native

English: Working proficiency